A) ANCHOR SCENES (3–6)

SCENE #1: Discovering Heat-Shock Genes (Grad School Breakthrough)
Summary: In graduate school, Lindquist uncovered that heat stress triggers specific gene activity. She demonstrated that brief heat shock in fruit-fly cells induces a new set of "heat-shock" proteins while normal protein synthesis halts. This tied mysterious chromosomal "puffs" (seen in Drosophila polytene chromosomes) to the production of those heat-shock proteins. Her 1975 experiments, conducted with advisor Matthew Meselson and colleague Steve Henikoff, proved that heat-shock puffs were sites of induced RNA transcription for these proteins.
Date/Place: ~1974–1975, Harvard University (Cambridge, MA) – findings presented at University of Chicago's Whitman Lab in 1974.
People: Susan Lindquist (then a PhD student), her mentor Matthew Meselson, labmate Steven Henikoff; Richard I. Morimoto (a fellow student who witnessed her results).
Concrete details: Morimoto recalls wandering into the lab and Lindquist eagerly showing him an "fascinating" result: after a heat shock, **Drosophila** cells "induce synthesis of a small set of proteins (heat-shock proteins) while repressing preexisting" proteins. She further showed the newly made heat-shock mRNAs hybridized to the chromosome "puffs" that appear during heat stress, directly linking those puffs to specific induced genes. Her first paper (PNAS 1975) documented this phenomenon. It was striking because many biologists had dismissed the heat-shock response as a lab oddity, but Lindquist's evidence suggested it was a fundamental genetic program.
Why it matters: This scene marks Lindquist's first major discovery and *established a new field* of stress biology. By proving that organisms activate dedicated genes under stress, she laid groundwork for cloning heat-shock genes and understanding protein chaperones. Her PhD work *inspired other scientists* (like Morimoto) to study cellular stress responses. It also foreshadowed Lindquist's career theme of tackling "curiosity-driven" questions that others overlooked.
Evidence quality: **Primary.** First-hand recollection by Morimoto (published in an academic obituary) and Lindquist's own thesis/paper provide direct documentation.
Myth risk: **Low.** The anecdote of her showing the result is documented by a peer; the significance of heat-shock proteins is supported by her 1975 paper. (If anything, early skepticism by others is confirmed in sources, not a later myth.)
Sources: Morimoto 2016 (Nat. Struct. Mol. Biol.); Hightower 2017 (Cell Stress Chaperones).

SCENE #2: Risking a Career Switch to Yeast Genetics
Summary: As a young faculty member, Lindquist made a daring decision to switch her research organism from fruit flies to yeast *before* earning tenure. In the late 1970s at University of Chicago, she learned of a revolutionary method to "knock out" yeast genes

(developed by researchers in Boston) and realized its power. Enthusiastic about genetic tools in yeast, she decided to pivot her entire research to this simpler organism. A senior colleague, yeast geneticist Rochelle Esposito, privately advised her that such a drastic change, especially for a junior professor, was **"really risky"** pre-tenure. At the time, female scientists received little mentoring, so Esposito's warning was well-intentioned. Nonetheless, Lindquist was undeterred. She felt *lucky* just to have a faculty position and told herself it was "a miracle" she was there at all – so she might as well pursue what excited her. She went ahead and refocused on yeast genetics, despite the potential career hazard.

Date/Place: ~1978, University of Chicago, Department of Molecular Genetics & Cell Biology.

People: Susan Lindquist (new assistant professor), Rochelle Esposito (yeast geneticist colleague).

Concrete details: Lindquist heard about a new gene deletion technique in yeast (from Orr-Weaver, Szostak, Rothstein) and thought, *"Wow…that is really powerful."* This realization occurred while she was still doing Drosophila cell culture work. Esposito pulled her aside in the hallway to caution that switching model organisms could jeopardize her tenure, something Esposito "didn't normally interfere" about. Lindquist recalls appreciating the advice but not feeling frightened by the risk. In her words, **"I thought it was a miracle that I got this faculty appointment… I just wanted to follow what was exciting for me."**. She proceeded to adopt *Saccharomyces cerevisiae* (baker's yeast) as her new experimental system.

Why it matters: This pivotal decision set the stage for Lindquist's most significant discoveries. Embracing yeast allowed her to probe protein folding and heredity with powerful genetics. It also illustrates her fearless, curiosity-driven style – willing to defy conventional career wisdom for the sake of science. The switch paid off: yeast became her lifelong model and led to breakthroughs in prions, chaperones, and disease models. Narratively, this scene highlights Lindquist's independence and willingness to take risks in science **even at a personal career cost**.

Evidence quality: **Primary.** Lindquist herself recounted this transition and conversation in an interview (The Scientist, 2016) with direct quotes.

Myth risk: **Low.** This is well-documented in her own words. (The notion that switching fields pre-tenure is risky is a general truth in academia, not a dramatic embellishment.)

Sources: Azvolinsky 2016 (The Scientist profile).

SCENE #3: Proving a Protein-Only Inheritance – the Yeast Prion

Summary: In the 1990s, Lindquist helped crack a long-standing mystery in yeast genetics by showing it was caused not by DNA, but by a misfolded protein "prion." A colleague, Dr. Yury Chernoff, had been studying an odd hereditary factor in yeast called *[PSI+]* that made cells read through stop codons (affecting protein translation

fidelity). Chernoff found that yeast cells needed a heat-shock protein (Hsp104) for *[PSI+]* to be inherited, which intrigued Lindquist since her lab had discovered Hsp104's role in dissolving protein aggregates. In 1994, NIH researcher Reed Wickner hypothesized that *[PSI+]* might be a prion – a self-perpetuating protein state – rather than a gene. Lindquist's team set out to test this bold idea. By 1996, **they demonstrated biochemically that the normal yeast Sup35 protein can flip into an aggregated, self-propagating conformation (the *[PSI+]* prion) which is passed from mother cell to daughter cell**, causing the heritable trait. This was strong evidence that a protein-only mechanism underlay the inheritance of *[PSI+]*.

Date/Place: 1994–1996, University of Chicago – culminating in a *Science* paper in 1996.

People: Susan Lindquist (principal investigator), Yury Chernoff (collaborator who first called her about *[PSI+]*), Reed Wickner (whose prion hypothesis guided the work), and Lindquist lab members (such as researcher Laura Li, co-author on 1996 paper).

Concrete details: Lindquist remembers **"a call from Yury Chernoff"** about a mysterious genetic factor affecting translation fidelity. Her lab had data that Hsp104 was somehow required to maintain that factor, but initial reports of Hsp104's aggregate-busting activity were met with skepticism ("no one would believe us – it was…very unexpected" she noted). When Wickner's 1994 paper proposed the factor was a prion, Lindquist immediately saw the connection. She worked with Chernoff to test if the factor was indeed a protein-based inheritance. They showed that in *[PSI+]* cells, Sup35 protein (a translation terminator) had aggregated into amyloid-like clumps. If they "cured" the yeast of these aggregates (by eliminating or inactivating Hsp104), the trait (read-through of stop codons) disappeared. The Sup35 protein could *switch* to a prion state that self-templates its misfolded form and is transmitted to cell progeny. Notably, the Sup35 protein's prion-form caused ribosomes to ignore stop signals, producing longer proteins than normal. This explained the odd genetic behavior purely through protein conformation. In 1996, Lindquist's group published "Support for the prion hypothesis for inheritance of a phenotypic trait in yeast," confirming that a new phenotype could be inherited via protein folding rather than DNA.

Why it matters: This was a **revolutionary moment** for molecular biology. It provided the first robust evidence that the Central Dogma (DNA→RNA→protein) has exceptions – information can be transmitted by protein shape alone. Lindquist's finding expanded the concept of heredity and helped establish **prions** as not just disease agents in animals but as possible epigenetic elements in lower organisms. Scientifically, it opened up an entirely new subfield on protein-based inheritance. For Lindquist's career, it cemented her reputation: she showed extraordinary insight in pursuing an idea that many found implausible.

Evidence quality: **Primary/Secondary mix.** Lindquist detailed the sequence of events in a PNAS Q&A (her own account). The outcome is recorded in her 1996 *Science* paper (a primary source referenced in retrospectives).

Myth risk: **Medium (managed).** The "eureka" narrative of discovering the *protein-only inheritance* is supported by firsthand reports, but one must remember she built on Wickner's hypothesis. We have flagged that Lindquist did not single-handedly "invent" prions – rather, she *proved* in yeast what others hypothesized. We present the scene with proper credit to avoid an oversimplified hero story.
Sources: Lindquist interview 2011 (PNAS); Goudsmit 2016 blog (citing *Science* 1996).

SCENE #4: A Prion in Memory? Collaboration with Kandel
Summary: A chance encounter between Lindquist and neuroscientist Eric Kandel led to a cross-disciplinary experiment showing that prion-like proteins might stabilize long-term memory. Around 2003, Kandel (a Nobel laureate for memory research) approached Lindquist with a curiosity: one of his key synaptic proteins, CPEB, contained a peculiar domain resembling a yeast prion sequence. He wondered if the protein, which is involved in maintaining neural synapses, could act like a prion in neurons. Lindquist agreed to help test this by leveraging her yeast system. Their teams collaborated to insert the Aplysia (sea slug) CPEB protein into yeast cells. Remarkably, they found that CPEB could indeed switch into a self-perpetuating aggregated form in yeast, analogous to a prion, which turned out to be the *active* form of the protein. This provided a model for how lasting memories might be encoded at the molecular level.
Date/Place: 2001–2003, Whitehead Institute, MIT (Cambridge, MA), culminating in a *Cell* paper published August 2003.
People: Susan Lindquist; Eric Kandel (Columbia Univ. neuroscientist); Kausik Si (Kandel's postdoc who identified the prion-like domain in CPEB); and Lindquist lab members assisting with yeast experiments.
Concrete details: Kandel literally **"walked into [Lindquist's] office one day, asking whether prions might be involved in memory"**. He brought the observation that the neuronal protein CPEB had a region strikingly similar to yeast prion proteins. Instead of dismissing the idea as far-fetched, Lindquist jumped in. Her lab took Kandel's CPEB and expressed it in yeast cells as a test system. They observed that CPEB underwent a **conformational switch** in yeast – forming aggregates – which **"rendered the protein active."** In yeast, that prion-state was self-perpetuating. Extrapolating to neurons, they proposed that when specific nerve cells are stimulated, CPEB in those synapses might convert to an aggregated state that *persists*. Lindquist explained that this could create a long-lasting molecular footprint—essentially a stable biochemical memory at the synapse. Further, once switched, the protein could adopt any of several stable activity states that last years, possibly correlating with memory strength. The findings, published in *Cell* (Si et al. 2003), suggested a novel mechanism for memory storage: a prion-like protein that keeps synapses persistently active.
Why it matters: This scene shows Lindquist taking her prion expertise into a completely new realm: neuroscience. The idea that a prion-like mechanism could underlie memory

was *audacious* and not widely accepted at first, yet their evidence was compelling. It mattered scientifically because it linked a molecular phenomenon (prions) to a higher-order function (long-term memory), bridging two disparate fields. For Lindquist's narrative, it highlights her collaborative spirit and intellectual fearlessness – she was willing to test a "crazy idea" from another field, and it paid off with a high-impact discovery. This collaboration also underscored the broader relevance of her prion research beyond yeast, influencing how neuroscientists think about memory mechanisms.
Evidence quality: **Primary.** Lindquist herself described the Kandel interaction and experiments in the PNAS Q&A. The result was published in a peer-reviewed journal (supporting source). Multiple retrospectives cite this story as well.
Myth risk: **Low.** The collaboration and its outcome are documented in scientific literature. The almost serendipitous story of Kandel walking into her office might seem "too perfect," but it comes from Lindquist's own account. We treat it as factual given the direct source.
Sources: Lindquist interview 2011 (PNAS); Goudsmit 2016 (recounting Kandel collaboration).

SCENE #5: Yeast to the Rescue – Tackling Neurodegenerative Diseases
Summary: In her late career, Lindquist turned her focus toward curing diseases by using yeast as an unexpected yet powerful model. Around 2003, she proposed that proteins involved in Parkinson's, Alzheimer's, ALS, etc., could be studied in simple yeast cells – an idea many found crazy. In fact, she admitted **"Even people in my lab thought we were crazy to try to study neurodegenerative diseases with a yeast cell. It's not a neuron."**. Nonetheless, she and her team persisted. They engineered yeast strains to produce human proteins that misfold in disease, such as α-synuclein (implicated in Parkinson's) and huntingtin (in Huntington's disease). Using these yeast models, Lindquist's lab performed massive genetic screens to find genes and compounds that affect toxicity. This approach—dubbed "humanized yeast" models—paid off. In 2003, they published in *Science* a set of yeast genes that, when mutated, worsened alpha-synuclein's toxic effects, revealing pathways of lipid metabolism and cell transport that were key to Parkinson's pathology. Encouraged by such successes, Lindquist co-founded biotech companies to translate these discoveries. In 2003 she co-founded **FoldRx Pharmaceuticals**, aiming to develop drugs for diseases of protein misfolding (FoldRx later developed tafamidis, a drug for a fatal amyloid disease). In 2014, despite battling illness, she helped launch **Yumanity Therapeutics**, focusing on Parkinson's and other neurodegenerative disease drug discovery. In her final years, she passionately drove these efforts to make sure the science of protein folding could lead to real treatments.
Date/Place: 2003–2016, Whitehead Institute (MIT) and Boston biotech startups (FoldRx in Cambridge, Yumanity in Cambridge/Boston).

People: Susan Lindquist; her postdocs and students (e.g., Tricia Serio, Aaron Gitler, Vikram Khurana among others); co-founders like Susan Lindquist's former student and colleagues at FoldRx, and Tony Coles (CEO of Yumanity).

Concrete details: Lindquist's lab showed tremendous determination in this period. For example, in one 2003 study they systematically knocked out each of ~4,800 yeast genes in cells producing toxic human proteins. They found dozens of yeast genes that, when missing, made α-synuclein toxicity worse, most of them related to lipid metabolism and vesicle traffic. These results gave *"proof of concept"* that yeast could reveal cellular mechanisms of human disease. Colleagues recall that Lindquist never wanted to do an experiment unless it could have deep biological or medical impact. True to that, she pushed beyond publishing papers: she translated findings into action. FoldRx (which she co-founded) discovered a drug that stabilizes a misfolded human protein (TTR) – that drug, tafamidis, was approved to treat transthyretin amyloidosis. In 2014, even after a cancer diagnosis, Lindquist joined forces to create Yumanity Therapeutics. Dr. Vikram Khurana of Yumanity noted, *"She did not want to do an experiment unless she thought it would ... make a difference to the world."* Lindquist remained scientifically active literally until she couldn't; her illness was kept private, and colleagues were stunned to hear of her passing during a conference. This scene, therefore, spans her "last experiment" – using yeast to conquer neurodegeneration – which was both a scientific and personal mission.

Why it matters: Lindquist's late-career pivot to disease research illustrates the *practical* power of fundamental biochemistry. She took an unlikely model (yeast) and showed it can illuminate complex human illnesses. This democratized the study of neurodegeneration, enabling high-throughput screens that were impossible in neurons. The outcome of this work has been new therapeutic strategies and even drug candidates for illnesses like Parkinson's. As a narrative capstone, it shows Lindquist's enduring drive to **"serve society"** with science – turning her decades of protein-folding expertise into hope for patients. It also provides a poignant final chapter: even as she faced mortality, she was setting the stage for future cures, embodying her belief that scientists have a duty to tackle big problems.

Evidence quality: **Primary and reputable secondary.** Lindquist's own quote of lab skepticism comes via *NY Times* (2007). The scientific results are documented in high-profile journal articles (Science 2003 etc.) and summarized in expert commentary. Obituaries and colleagues (MIT News, Boston Globe) confirm her role in founding companies and her mindset about impactful science.

Myth risk: **Low.** Some might frame her as single-handedly "finding a cure" – we avoid that. We document the stepwise progress (e.g. 2003 study) and note that these were ongoing efforts. The portrayal of her as working up to her last days is based on colleagues' accounts, which we treat carefully (it's true she kept contributing until very late). Overall, this scene is supported by concrete records and first-person quotes.

Sources: Science of Parkinson's blog (quoting NYT 2007); MIT News 2016; Boston Globe via MIT; Hightower 2017.

B) DIRECT QUOTES (3–7)

- "I thought, 'Wow, you could knock out genes; that is really powerful.' So I decided I would work on yeast."
  Context/date: Recalling her decision in the late 1970s to switch from fruit flies to yeast research, after learning of new gene knockout methods. Interview with *The Scientist*, published January 2016.
  Source: Anna Azvolinsky, *The Scientist* (2016).
- "I thought it was a miracle that I got this faculty appointment and was so happy to be there for a few years that I just wanted to follow what was exciting for me."
  Context/date: Explaining why she wasn't afraid to take risks early in her career (in the late '70s), since simply having a faculty position felt miraculous. From the same 2016 interview, reflecting on advice she received as a junior professor.
  Source: Anna Azvolinsky, *The Scientist* (2016).
- "Yes, we now know there are lots of these prions; and flipping on the switch to a prion state can turn the function of the protein on or off and potentially provide yeast cells with novel, heritable traits."
  Context/date: Describing the basic idea of **beneficial prions** – how misfolding a protein can create a heritable switch – in a Q&A interview. Published in *PNAS* on Dec 13, 2011. Lindquist was addressing how a prion can sometimes *benefit* an organism by conferring new traits.
  Source: Prashant Nair, "QnAs with Susan L. Lindquist," *PNAS* 108(50): 19861 (2011).
- "Prions are fundamentally a protein folding problem. Every once in a while, cells might switch some of their proteins into and out of the prion state, just out of happenstance. We call this a 'bet-hedging' strategy; but stress and unfavorable environments cause protein folding problems and increase the frequency of that switch. It's an illustration of environmental change altering the pace of evolution. It's very Lamarckian: Heritable traits arise as adaptations before they are hardwired into the genome."
  Context/date: Lindquist's explanation of what triggers a prion switch and its evolutionary significance, from the same *PNAS* Q&A (2011). Notably, she likens the process to a *"bet-hedging"* mechanism and even references Lamarckian evolution to describe how acquired traits can appear before genetic changes occur. This encapsulates her forward-thinking perspective on evolution and protein folding.
  Source: Prashant Nair, *PNAS* interview (2011).

- "Even people in my laboratory thought we were crazy to try to study neurodegenerative diseases with a yeast cell. It's not a neuron."
  Context/date: Recalling the skepticism she faced in the early 2000s when she proposed using yeast as a model for brain diseases like Parkinson's. Quoted in *The New York Times* (2007). This statement shows her awareness that the idea sounded wild, yet underscores her visionary willingness to pursue it.
  Source: Quoted by Nicholas Wade, *The New York Times* (July 2007), via *Science of Parkinson's* blog.
- "It has been said that you can't understand biology until you understand evolution. I would argue that you can't understand evolution or biology until you understand **protein**."
  Context/date: Interview remark circa 2002, when Lindquist was named one of the "50 Most Important Women in Science." She was emphasizing the central role of proteins (and protein folding) in life's processes. Published in *Discover Magazine* (Nov 2002 issue).
  Source: Kathy Svitil et al., *Discover* 23(11): "The 50 Most Important Women in Science" (2002).

C) HUMAN REALITY DETAILS (8–15 bullets)

- Lindquist was born in 1949 on Chicago's South Side, the daughter of a Swedish-American father and Italian-American mother; neither parent had a college degree, but they strongly valued education. Her upbringing instilled a drive to learn. — *Source: Shorter 2017, TiBS*.
- By fifth grade (around age 10), she became *"enthralled by the question: what is life?"* – an unusually early passion for fundamental biology that she carried into her career. — *Source: Shorter 2017, quoting family recollections*.
- She earned her B.A. in microbiology at the University of Illinois Urbana-Champaign in 1971 (supported by a NSF Fellowship) and then a Ph.D. in biology from Harvard in 1976. At Harvard, she joined Matthew Meselson's lab and began probing how cells respond to heat stress. — *Sources: Shorter 2017*; *Hightower 2017*.
- For her doctoral thesis, Lindquist studied **protein and RNA synthesis induced by heat treatment** in fruit flies. This was a risky topic at the time – many senior scientists dismissed the heat-shock response as a mere "curiosity," but Lindquist forged ahead and proved it was a real biological phenomenon. — *Sources: Hightower 2017*.
- She did a brief postdoc at the University of Chicago with cell biologist Hewson Swift, and was soon recruited to Chicago's faculty in 1978. She spent 23 years at U. Chicago, rising to full Professor and becoming an HHMI Investigator and the

Albert Lasker Professor of Medical Sciences. In 2001, she moved to MIT to join the Whitehead Institute, which she directed from 2001–2004. — *Sources: Morimoto 2016*; *Shorter 2017*.

- During her Chicago years, she married **Edward ("Ed") Buckbee**, a scholar of French medieval literature, and they had two daughters, Alana and Nora. Friends recall that she balanced family and science with help from Ed – and she remained very close to her family throughout her career. — *Source: Shorter 2017*.
- Lindquist was *"not a geneticist, not a biochemist, not a molecular biologist"* exclusively – **she was fond of saying that she used *all* approaches necessary to answer the question at hand**. Indeed, her work ranged from genetics to cell biology to biophysics. This open-minded style helped her make interdisciplinary leaps (e.g. from yeast to neuroscience). — *Source: Morimoto 2016*.
- Colleagues and trainees describe Lindquist as having an **"infectious esprit"** for discovery paired with *disarming warmth and generosity*. She created an incredibly **nurturing lab environment**, attracting scientists from diverse backgrounds (physicists, chemists, physicians – everyone was welcome in her lab).
- As a mentor, she was *"inspirational, nurturing, and indefatigable."* She trained over a hundred students and postdocs; at MIT alone (2001–2016) she supervised **115 researchers-in-training** (from undergrads to fellows). Her trainees have often remarked on how much time and energy she invested in their development. — *Sources: Shorter 2017*; *MIT News 2016*.
- Lindquist had a *hands-on* and Nurturing leadership style. She would sometimes invite her whole lab to her home on weekends to work on paper manuscripts together over coffee, treating her students like extended family (they got to know her husband and kids). She also made time for play: despite a packed schedule, she would sit on the grass outside the Whitehead Institute on Fridays to **watch her students' weekly soccer games**, cheering them on. These personal touches endeared her to her lab and solidified a strong team spirit. — *Source: Shorter 2017 (memorial anecdotes)*.
- Known for clarity of thought, Lindquist wrote scientific papers that were **unusually accessible** and gave seminar talks that were *captivating* even to non-specialists. One former student (James Shorter) recalls being so inspired by her seminar at a 1998 conference in London that he "was inspired to postdoc in her laboratory" immediately after his Ph.D..
- She was also a passionate advocate for diversity in science. As one of the few prominent women in molecular biology of her era, Lindquist actively mentored other women. At Whitehead, she launched a fund to encourage women in science, and she often spoke out about improving science education (even

praising countries like Singapore for valuing early education). Tyler Jacks of MIT's Koch Institute noted, *"Susan was a tireless advocate for women in STEM… a role model for women in science."*. — *Sources: Koch Institute memorial 2016*; *Lake Forest talk 2007*.

- Honors poured in for her groundbreaking work: she was elected to the National Academy of Sciences in 1997, received the National Medal of Science in 2010 (presented by President Obama), and was one of very few scientists elected to Britain's Royal Society (2015) while still actively running a lab. Despite accolades, colleagues say she remained down-to-earth and more excited about new data than about awards. — *Source: Shorter 2017*.
- **Final days:** Lindquist was privately battling cancer in 2015–2016. She largely kept her illness to herself, so the scientific community was stunned when she passed away on October 27, 2016. True to form, she had "fought to the last moment," remaining intellectually active. For instance, she continued mentoring and strategizing research at Whitehead as long as she could. When news of her death reached a 2016 heat-shock protein conference, many there had not even known she was sick – a testament to her strength and absence of self-pity. Friends and colleagues held an impromptu champagne toast at the meeting to honor her life in science. — *Source: Hightower 2017*.

D) SCIENCE CONNECTION (Intro Biochem)

- **Best-fit topic:** **Protein folding and molecular chaperones.** (Also relevant to DNA→RNA→Protein information flow, i.e. the Central Dogma, and its exceptions.)
- **Precise contribution:** Susan Lindquist showed that **alternative protein conformations can act as heritable elements** in cells. She discovered that misfolded proteins called *prions* can transmit information across generations without DNA changes. She also elucidated the role of chaperone proteins (like Hsp90 and Hsp104) in stabilizing proteins and buffering genetic variation.
- **Bridge explanation:** In biochemistry class, students learn that a protein's function depends on its 3D structure (fold). Lindquist's work is a dramatic illustration of this principle. She found that when a protein folds into an **amyloid** or prion form, it can **change the phenotype** of the organism by altering what proteins get made or how they act. For example, in yeast the Sup35 protein normally helps stop protein synthesis at the end of mRNA (terminating translation). But in its prion form, Sup35 clumps up and can't do its job, so ribosomes ignore stop codons and produce extra-long proteins. This "read-through" effect gives the yeast new traits (sometimes helpful ones) without any mutation in the DNA – the change is purely at the protein level. Lindquist also introduced the concept of **molecular chaperones** as evolutionary "capacitors."

Chaperones like Hsp90 help other proteins fold correctly; Lindquist showed that Hsp90 can **mask genetic mutations' effects** under normal conditions, but under stress (when Hsp90 is busy or inactivated), hidden mutations suddenly express new traits. This means stress can reveal variation that was silent, potentially accelerating evolution. All these discoveries tie directly to fundamental biochemistry: protein structure, folding, and the regulation of protein function in the cell.

- **Scientist's own framing/analogy:** Lindquist often likened prions to an evolutionary **"bet-hedging"** mechanism. She explained that a cell occasionally (and especially under stress) flips certain proteins into a prion state, essentially *rolling the dice* to see if a new trait might help it survive. If the trait is beneficial, the prion state can be passed on to offspring – a bit like **Lamarckian inheritance**, she said, where an organism adapts to the environment first and the genome catches up later. This vivid analogy helps students connect protein folding with evolutionary theory. (She coined the term "**protein-only inheritance**" to describe this concept.)
- **Sources:** Lindquist Q&A – *PNAS* 2011; MIT News obit 2016; Morimoto 2016 (on Hsp90 as capacitor); Shorter 2017 (TiBS, on Hsp90).

E) COMPLICATIONS OR CONTROVERSIES

- **Prions: Deadly menace or evolutionary tool?** There was debate over Lindquist's claim that prions can play *positive* roles in biology.
  - **What's agreed upon:** Prions were first notorious for causing diseases (e.g., mad cow disease). Lindquist's yeast studies, however, showed that misfolded proteins could also **confer beneficial traits**. In 2012, her lab reported that roughly one-third of wild yeast strains harbor prion-based traits – some making yeast more fit in stressful environments. For instance, a wine yeast prion made cells resistant to an antifungal drug, and losing the prion lost that resistance. This supported the idea that prions might serve as an adaptive "switch" that yeast can flip to survive changing conditions.
  - **What's disputed:** Many biologists remained skeptical of generalizing this as an evolutionary strategy. Some argued that calling these proteins "prions" stretches the definition, or that the frequency and long-term benefit of prion switching in nature were not yet proven. Even Lindquist acknowledged in 2011, **"It is still a fairly controversial idea. Some people believe it, and others don't."**. Critics pointed out that while 25% of prion switches were beneficial in her lab conditions, many were neutral or harmful – so prions might be an evolutionary gamble more than a

designed tool. The very term "Lamarckian" raised eyebrows, as it challenges orthodox evolutionary theory.

- **Sources (both sides):** Lindquist's perspective in *PNAS Q&A* (2011); Jaap Goudsmit blog (2016) describing her 2012 Nature findings. These highlight the promise of prions in evolution, but also note ongoing debate. (No open feud ensued – it was a scientific discourse on interpretation.)

(Overall, Lindquist's career was relatively free of personal controversy. She was widely respected, and most "conflict" came from her willingness to embrace unorthodox ideas that initially met skepticism. In the end, many of her once-controversial ideas – like protein-only inheritance – gained broad acceptance.)

F) "NOT ON WIKIPEDIA" FINDS

- *Heat-shock heresy:* As a student, Lindquist chose to study the heat-shock response at a time when many biologists dismissed it as *"a laboratory artifact… a curiosity of Drosophila biology."* This contrarian choice (inspired by a tip from a junior faculty member) showed her early boldness. *Why distinctive:* It reveals how she gravitated to big questions others ignored, a pattern seen throughout her life. — *Source: Hightower 2017*.
- *Lindquist McKenzie:* In the mid-1970s, Susan went by **"Susan Lindquist McKenzie."** She published her 1975 PNAS paper under that name. *Why distinctive:* This little-known fact (from her thesis era) reflects a brief first marriage or name change that isn't commonly noted. She later published as Susan Lindquist, which is how history remembers her. — *Source: Morimoto 2016*.
- *Tenure risk-taker:* Before tenure, she defied conventional wisdom by switching model organisms (from flies to yeast) and research direction. A senior colleague warned her against it, but she went ahead, saying **"I just wanted to follow what was exciting."**. *Why distinctive:* Illustrates her fearless attitude and priority for science over job security – a biographical detail often glossed over in favor of her later achievements. — *Source: The Scientist interview 2016*.
- *Protein evangelist:* Lindquist famously argued that understanding proteins is the key to understanding life. **"You can't understand evolution or biology until you understand protein,"** she said. *Why distinctive:* This quote captures her philosophy in a nutshell. It doesn't appear in basic biographies but is telling of how she viewed her work's importance in the grand scheme. — *Source: Discover Magazine 2002*.
- *Lab as family:* She blurred the lines between work and home. **On weekends, Lindquist would invite lab members to her house** to jointly write papers and brainstorm. She'd cook or order food, making it a comfortable, collaborative

space. She also took joy in her students' extracurriculars – for example, cheering on **lab soccer games** every Friday after work on MIT's lawns. *Why distinctive:* These anecdotes show a human side of Lindquist – a mentor who cared personally for her team – which isn't captured in formal profiles. It helps readers picture her not just as a scientist but as a person and leader. — *Source: Shorter 2017 (memorial tribute)*.

- *Stealth humanitarian:* Beyond academia, Lindquist co-founded at least **two companies** (FoldRx and Yumanity) to translate her discoveries into cures. While Wikipedia might list the companies, it doesn't convey the personal motivation: she felt a *"moral obligation"* for scientists to use their talents for societal good. She often said researchers must reach beyond knowledge for its own sake and strive for impact. *Why distinctive:* It highlights her sense of duty and the late-career shift to medical innovation, driven by her conscience – a narrative nuance beyond a CV line. — *Sources: MIT News (David Page quote)*; *Science of Parkinson's blog*.
- *Quiet courage:* In 2015–2016, Lindquist chose to keep her battle with cancer largely private. She continued working as long as possible, not wanting pity or reduced expectations. When she died, many colleagues were surprised to learn how gravely ill she had been. *Why distinctive:* This speaks to her character – stoic and committed – and it's a touching detail that goes beyond the typical professional bio. — *Source: Hightower 2017*.

G) SOURCES CONSULTED (Full citations)

Interviews and Oral Histories:

- **Nair, Prashant.** (2011). *QnAs with Susan L. Lindquist.* **Proceedings of the National Academy of Sciences, 108**(50), 19861. DOI: 10.1073/pnas.1118394108. (Transcript of a PNAS interview in which Lindquist discusses prions, Hsp104, and more in her own words.)
- **Azvolinsky, Anna.** (2016, January). "Fearless about Folding." *The Scientist.* (Profile piece on Lindquist's career; contains direct quotes about switching to yeast and taking risks.)
- **Science of Parkinson's blog (Simon Stott).** (2017, March 8). "A yeast model of Parkinson's disease." *(Blog post).* (Memorial blog explaining Lindquist's yeast approach to neurodegeneration, with a quote from Lindquist's 2007 NYT interview about being thought "crazy" and details of her 2003 Science paper.)

Obituaries and Tributes:

- **Shorter, James.** (2017). "Susan Lee Lindquist (1949–2016)." *Trends in Biochemical Sciences, 42*(3), 169–171. DOI: 10.1016/j.tibs.2016.12.002. (A two-page in memoriam by a former postdoc; rich in biographical details and personal anecdotes.)
- **Morimoto, Richard I.** (2016). "Susan Lindquist 1949–2016." *Nature Structural & Molecular Biology, 23*(12), 1072–1073. DOI: 10.1038/nsmb.3339. (Obituary by a colleague, recounts her PhD discovery and her mantra of using all systems in science.)
- **Hightower, Lawrence E.** (2017). "Susan Lee Lindquist (1949–2016): It is personal." *Cell Stress & Chaperones, 22*(1), 1–4. DOI: 10.1007/s12192-016-0759-4. (Editorial tribute; provides historical context on heat shock research and a first-person account of hearing of her passing at a conference.)
- **MIT News (Whitehead Institute).** (2016, Oct 28). "Susan Lindquist, pioneering biologist and former director of Whitehead Institute, dies at 67." *MIT News Office.* (Official obituary/news release with quotes from colleagues and summary of her work on prions and diseases.)
- **Koch Institute at MIT.** (2016, Oct 28). "Remembering Scientific Pioneer Susan Lindquist." (Short memorial announcement with a quote from Tyler Jacks regarding her mentorship and advocacy.)
- **Goudsmit, Jaap.** (2016, Nov 7). "Susan Lindquist (1949–2016): A daring voice of science silenced far too early." *(Personal blog post.)* (Recollections by virologist Jaap Goudsmit, describing discussions with Lindquist on prions and summarizing her major prion papers. Although a blog, it cites specific studies and provides a colleague's perspective.)

Published Talks and Features:

- **Discover Magazine.** (2002, November). "The 50 Most Important Women in Science." *Discover, 23*(11), 52–58. (Feature article where Lindquist is profiled; includes her quote about understanding protein to understand life.)
- **Eukaryon (Lake Forest College).** (2008). "A Day with a Leading Scientist: Susan Lindquist" by Lokesh Kukreja. *Eukaryon, 4*, 7–8. (Student-written summary of Lindquist's 2007 lecture; provides a few direct quotes illustrating her excitement about science. *Note*: Undergraduate publication, not peer-reviewed.)
- **The Boston Globe.** (2016, Nov 3). "MIT's Susan Lindquist, 67, a pioneer in science." by Bryan Marquard. (Globe obituary quoting Dr. Vikram Khurana on her philosophy of only doing impactful experiments.)
- **The New York Times.** (2016, Oct 31). "Susan Lindquist, Trailblazing Molecular Biologist, Dies at 67." by William Grimes. (NYT obituary; mentions her concept of

proteins carrying inheritance and includes the "can't understand evolution until you understand protein" quote paraphrased.)

(Additional sources consulted include academic profiles, awards citations, and Lindquist's own scientific publications for verification of specific claims, such as the 1996 Science paper on [PSI+] and the 2012 Nature paper on wild yeast prions.)

H) FLAGGED "MYTH-RISK" DETAILS

- **Claim:** *"Prions are a deliberate evolutionary strategy that organisms use to adapt."*
  Where it appears: In some popular accounts of Lindquist's work and even in her own analogies, prions are portrayed as a bet-hedging tool that cells *intentionally* use to evolve new traits (e.g., references to Lamarckian evolution in media coverage). For instance, headlines after her 2012 paper suggested yeast prions are an "evolutionary memory" mechanism pre-designed for stress adaptation.
  Why sourcing is weak: While Lindquist's experiments showed **prions can confer adaptive traits** and arise more often under stress, there is no direct evidence that this is an evolved *purposeful* strategy versus a fortuitous byproduct of protein misfolding. The idea of nature "intentionally" using prions veers into teleology. Lindquist herself used cautious language like "bet-hedging" and acknowledged the controversy, but secondary reports sometimes overstated it as fact.
  Best available evidence: Lindquist's 2012 *Nature* study documented many beneficial prion traits in wild yeasts, suggesting prions can be evolutionary relevant. She and colleagues argued these traits indicate prions *could* be a means of fast adaptation. However, as she noted, some scientists remained unconvinced. Thus, the safest interpretation is that prions offer a *possible* evolutionary advantage under certain conditions – a fascinating hypothesis, but not definitively proven as a general strategy in nature. We present it with that nuance to avoid a simplistic "prions = evolutionary plan" narrative.

LENS RECOMMENDATION

Lens scoring (0–3):

- **Eureka/Discovery:** **3** – Lindquist's story features a well-documented "Eureka" in proving protein-only inheritance. The [PSI+] prion discovery scene (Scene 3) is vividly documented with primary sources, making a discovery-focused narrative very strong.

- **Failure/Setback:** **2** – There is a theme of overcoming skepticism and initial failures (e.g. trouble publishing the Hsp104 result, warnings about switching fields). These struggles exist but were ultimately turning points rather than prolonged failures, so they support a narrative of persistence, albeit with moderate detail.
- **Mentorship/Relationship:** **1** – No singular mentor-mentee transformative moment stands out. (She had great mentors like Meselson, and she *was* a great mentor, but there isn't a clear pivotal mentor scene in the evidence beyond general influence.)
- **Controversy:** **1** – Apart from scientific debate over prions, Lindquist had no major personal controversies. The prion controversy is one-sided in her favor (she proved her point, some remained skeptical). This would be hard to center a full narrative on.
- **Human Detail:** **2** – Her personal qualities (fearless risk-taker, collaborative leader, balancing family, etc.) are well-supported and could enrich a story. They provide color and motivation but would need to be tied back to scientific events to drive a narrative.
- **Last Experiment/Late career:** **2** – Lindquist's final chapter (applying science to neuro-diseases) offers a compelling wrap-up with real-world impact. It is documented and provides closure (her work lives on in companies), but it's more of an epilogue than a single dramatic climax.

Primary recommended lens: ** **Eureka – The "Protein-Only Inheritance" Discovery.
Why it fits: Lindquist's defining moment was her *paradigm-shifting discovery* that a protein can transmit heritable traits. This Eureka lens is supported by multiple anchor scenes: her heat-shock finding (Scene 1) sets the stage for thinking outside the box, and the prion breakthrough (Scene 3) is the narrative centerpiece, complete with skeptics, a "mysterious factor," and the thrill of proof. Scene 4 (Kandel collaboration) even shows her extending that discovery to memory, underscoring its significance beyond yeast. These scenes provide the classic "aha!" moments and turning points ideal for a discovery-focused story.
Best opening scene: *Scene 3,* depicting the prion discovery, would grab readers immediately – e.g. **Lindquist picking up Chernoff's phone call about weird results, then conducting the experiment that **"showed…this inherited trait was caused by a protein aggregate"**. That dramatic revelation sets the hook. We might then flash back to Scene 1 or 2 to show how she was prepared for this Eureka by earlier choices.
One quote to use early: Quote #4, where she says *"Prions are fundamentally a protein folding problem… It's very Lamarckian: Heritable traits arise as adaptations before they are hardwired into the genome."*. This captures her visionary thinking at the moment of discovery and can foreshadow the implications of the Eureka.

Cautions: We must ensure credit is shared (Wickner's role in hypothesizing the prion will be noted, to avoid the "solo genius" myth). Also, the "prions help evolution" idea should be presented as her hypothesis with evidence, while acknowledging it was controversial. But as a lens, Eureka works strongly – the sources allow us to describe the lab scene and Lindquist's excitement in credible detail.

Backup lens: "Overcoming the Impossible" – A Resilience/Skepticism lens (Failure-to-Success).
(Within 1 point of Eureka) This lens would focus on how Lindquist repeatedly tackled problems deemed "too risky" or "too unconventional" – and turned them into triumphs. It scores lower than Eureka only because no single failure defines her, but the *pattern* is compelling: She proved the worth of heat-shock research when others shrugged it off; she switched fields against advice; her prion theory faced disbelief until she delivered proof; she insisted on using yeast to study human disease despite "crazy" looks – and she was vindicated each time.

- **Scenes supporting it:** Scene 2 (mentor warning her about switching to yeast) and Scene 5 (colleagues thinking yeast for neurodegeneration was nuts) show external skepticism and potential failure points. Scene 3 (prion discovery) and even Scene 1 (heat shock courage) then show how her persistence paid off. These can be strung into a narrative of *defying the odds*.
- **Opening scene:** *Scene 2* might be a great start: young Sue Lindquist being told "Don't do this crazy thing" in 1978 – and her inner voice saying *"it's a miracle I'm here, I'll do what excites me"*. This immediately establishes the conflict (risk vs. reward) and her character.
- **Quote early on:** Quote #5, *"Even people in my lab thought we were crazy to try… It's not a neuron,"* would highlight the theme of facing doubt; or Quote #2 *"I thought it was a miracle…just wanted to follow what was exciting"* to show her mindset when confronting risk.
- **Cautions:** This lens shouldn't drift into portraying her as constantly embattled or an underdog – in truth, she gained recognition relatively early. The skepticism she faced was real but often short-lived until she proved her point. We'll need to emphasize her confidence and the soundness of her science, so it doesn't seem like contrarianism for its own sake. When well-framed, this "resilience" lens can be very inspiring, and it naturally integrates many stages of her career.

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